

Algebra II | Guided notes — we fill these in together

Vocabulary

Parabola: The U-shaped graph produced by a quadratic function.

Vertex: The turning point of a parabola, written as the ordered pair (h, k) .

Vertex Form: A quadratic equation written as $y = a(x - h)^2 + k$, where (h, k) is the vertex.

We do together. Identify a , the vertex, and the direction of opening for $y = 2(x - 3)^2 + 1$.

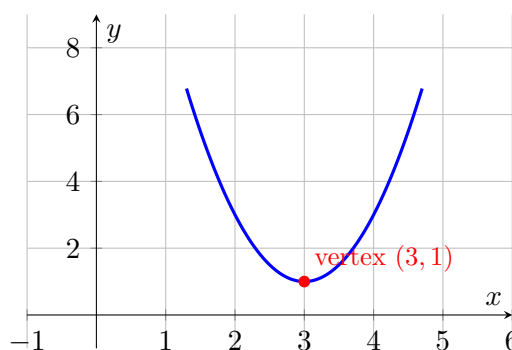
Step 1. Match the equation to $y = a(x - h)^2 + k$: here $a = \underline{2}$.

Step 2. Since the equation has $(x - 3)$, we know $h = \underline{3}$.

Step 3. The constant added at the end gives $k = \underline{1}$.

Step 4. So the vertex $(h, k) = \underline{(3, 1)}$.

Step 5. Since $a = 2$ is positive, the parabola opens up.



$y = 2(x - 3)^2 + 1$ opens up with vertex at $(3, 1)$.

You Try. Solve each. Show every step, then name the answer.

a) Identify a , the vertex, and direction of opening for $y = -(x + 4)^2 + 2$. $a = -1$, vertex $(-4, 2)$, opens down.

b) Identify a , the vertex, and direction of opening for $y = \frac{1}{2}(x - 5)^2 - 3$. $a = \frac{1}{2}$, vertex $(5, -3)$, opens up (wider than $y = x^2$).

Summary (fill in): In vertex form $y = a(x - h)^2 + k$, the vertex is always (h, k) , remembering that the sign of h inside the equation is the opposite of the value you write in the ordered pair. The sign of a tells direction: positive means opens up, negative means opens down.